PRODUCT REQUIREMENTS DOCUMENT

Telecom Complaint Intelligence & Automated Resolution Assistant (TelConnect)

Professional, Customer-First Product Specification & System Architecture

Cognizant Hackathon • Use Case 13

Document	Specification (v7.0 Implemented Platform)
Version	7.0 — Autonomous Multi-Task Agent & Multilingual Intelligence PRD
Product Type	AI-powered telecom complaint intelligence, autonomous resolution & operations platform
Primary User	Telecom Customer (Mobile Data, Broadband, Fiber, Landline)
Operational Users	Support Desk, RF Engineers, Network Ops, Field Dispatch & Admin Teams
Agent Orchestration	LangGraph Multi-Task StateGraph (6-node state machine with dynamic tool execution)
Multilingual Engine	Hugging Face DistilBERT (multilingual-cased) + Groq Zero-Shot Neural Translation + Rule Fallback
Primary AI Provider	Groq API (Llama-3 models + Whisper STT) with complete deterministic offline fallback
Voice Stack	Groq Whisper (Multilingual Voice STT) + Browser Web Speech API (Bilingual TTS Readout)
Core Data Store	SQLite (tci.db with WAL mode, foreign keys, and immutable audit logs)
Knowledge Store	TF-IDF Vector Space / In-Memory Cosine Similarity Engine (scikit-learn)
Backend Framework	Python + FastAPI + Uvicorn (REST API + Background Pipeline Scheduler)
UI & Experience	React 18 + Vite SPA with Modern 40/60 Split Layout, Leaflet Heatmap & Recharts
Test & KPI Status	100% Pass Rate (49 / 49 Automated Pytest Unit & E2E Test Cases Passing)

Product Vision: Turn telecom complaints from isolated support tickets into a closed-loop intelligence system that can understand the customer in any language, verify issues with live line diagnostics, resolve eligible problems using grounded SOPs, track unresolved cases transparently, detect mass geographic incidents, and proactively notify affected customers to prevent repeated complaints.

1. Product Overview & Core Principles

The Telecom Complaint Intelligence & Automated Resolution Assistant is a customer-facing AI service backed by an operational support and network analytics platform. It accepts complaints through conversational text or voice, auto-detects and normalizes multilingual input, performs dynamic network line diagnostics, matches against active geo-incidents, retrieves approved troubleshooting guidance via RAG, creates priority-scored tickets, and verifies resolution satisfaction before closing.

1.1 Key Implemented Principles

- **Customer First & Closed Loop:** The workflow begins with the customer's problem and ends only when the customer explicitly confirms the issue is fixed or rates the resolution.
- **One Source of Truth:** Both the customer assistant and admin operations console interact directly through the same FastAPI backend and unified SQLite database (tci.db).
- **Autonomous Agentic Execution:** Driven by a LangGraph StateGraph where LLMs perform reasoning and synthesis while deterministic APIs handle privileged database mutations.
- **Multilingual & Transliteration Aware:** Pure bidirectional English and Hindi support powered by Hugging Face DistilBERT tokenization and Groq zero-shot translation.

• **Explainable Intelligence:** Every ML classification, priority score (P1–P4), and root-cause hypothesis exposes its underlying contributing factors rather than opaque black-box numbers.

2. Objectives & Success Criteria

Objective	Expected Outcome & Implemented Capability
Multilingual Understanding	Hugging Face DistilBERT + Groq normalization across English, Hindi (Devanagari), and Hinglish.
Network Line Diagnostics	Real-time telemetry measuring download/upload speed, ping latency, jitter, and packet loss.
First-Contact Resolution	Immediate intercept via active incident linkage and RAG troubleshooting SOPs before creating tickets.
Complaint Lifecycle	Full state machine: verification, ticket creation, SLA assignment, confirmation, reopening, escalation.
Status Transparency	Live status, assigned team, SLA countdown, and root-cause explanations delivered directly in chat.
Mass Incident Detection	Rolling baseline anomaly detector flagging regional complaint spikes and auto-opening incidents.
Root-Cause Investigation	AI dossiers synthesizing temporal, spatial, and symptom patterns with ≥ 3 checkable evidence bullets.
Proactive Notifications	Human-in-the-loop review queue for drafting and approving broadcast alerts to affected customers.

3. Users, Roles & Access Control

• **Customer Role:** Accesses the modernized 40/60 split chat interface, voice STT/TTS, line diagnostic testing, personal ticket tracking, and confirm-to-close workflow. Isolated from admin interfaces.
• **Support Admin & Network Ops:** Accesses universal CSV schema-mapping ingest, priority queue workbench, Leaflet regional heatmap, incident dossiers, alert inbox, notification queue, and audit logs.

4. Customer Journey & State Machine Model

The customer experience follows a 10-stage verified workflow: **Start** (Chat/Voice) → **Normalize** (DistilBERT) → **Diagnose** (Line Speed Test / Incident Check) → **RAG SOP Guidance** → **Resolution Attempt** → **Customer Confirmation** → **Ticket Creation** (P1–P4 SLA) → **Live Status Tracking** → **Reopen / Escalation** → **Closure & CSAT Rating**.

Deterministic Status Transitions: new → in_progress → waiting_for_customer → resolved_pending_confirmation → closed (Alternative paths: reopened, escalated). Every transition writes an immutable audit record to complaint_status_history.

5. Autonomous Customer Assistant (LangGraph Architecture)

The customer assistant is implemented as a 6-node **LangGraph StateGraph** workflow (backend/app/agent_graph.py):

Node Name	Input / State	Node Operation & Output Action
1. translate_input	Raw User Message	Detects language, runs Hugging Face DistilBERT tokenizer, and normalizes Hinglish/Hindi idioms to canonical English semantics.
2. route_intent	Normalized English	Evaluates multi-turn state continuations (confirming registration, fix feedback, ratings) and classifies conversational intent across 15+ paths.
3. retrieve_context	Customer Profile + Intent	Queries active area incidents in customer region and retrieves top-2 matching SOPs/resolved cases via TF-IDF cosine similarity.
4. execute_action	Intent + Context	Executes dynamic tools: line speed tests, ticket creation (register_complaint), status changes, or feedback recording. Compiles verified facts.
5. synthesize_response	Verified Facts + Profile	Invokes Groq Llama-3 with strict factual grounding rules. Strictly prohibits hallucinating ticket IDs, policies, or ETAs.
6. translate_output	Grounded Response	Ensures Hindi outputs render in authentic Devanagari script and prepares suggestion chips and diagnostic telemetry cards.

6. Dynamic Network & Line Diagnostics Tooling

Customers can request on-demand line tests directly via chat (e.g. *'run a speed test'* or clicking the **Speed Test** action tile). The system generates dynamic telemetry evaluating ping latency, jitter, packet loss, download speed, and upload speed based on the user's circle/region and service type (Fiber/Broadband vs Mobile Data), automatically linking to active area incidents when degraded.

7. Admin Operations Control Plane (16 Modules)

The Admin Console comprises 16 operational modules: **(1) Universal CSV Upload Wizard** with automatic column mapping; **(2) Overview Dashboard** with volume/sentiment charts; **(3) Priority Queue** with multi-facet filters; **(4) Ticket Management Drawer** with factor breakdowns; **(5) Team Assignment**; **(6) Resolution Proposer**; **(7) Status History Timeline**; **(8) Leaflet Regional Heatmap**; **(9) Spike Detection Engine**; **(10) AI Root-Cause Dossiers** with evidence signals; **(11) Incident Operations**; **(12) Real-Time Alert Inbox**; **(13) Proactive Notify Queue**; **(14) Executive Analytics**; **(15) Live CSV Export**; and **(16) Immutable Audit Logging**.

8. Complaint Intelligence & Priority Scoring Engine

Complaints are classified using TF-IDF feature extraction and Logistic Regression (Macro-F1 ≥ 80%). Priority scores (0-100) map to P1-P4 bands based on urgency, negative sentiment, churn/escalation risk, active incident bonus (+20), and repeat count. Every score displays its contributing factor chips in the admin queue for full transparency.

9. Dynamic Database Schema (SQLite tci.db)

Core Entities: users (role, credentials, region, service), complaints (complaint_id, priority, sentiment, SLA, status, incident_id), complaint_status_history (immutable audit of all state transitions), incidents (region, spike multiplier, root cause, evidence JSON), notifications (draft text, match reason, approval status), chat_messages (role, text, metadata JSON), resolutions (proposed fix, outcome), feedback (1-5 star ratings), and audit_logs (action traces).

10. Backend REST API Specification

Endpoint	Method	Role / Scope	Description & Output
/api/auth/signup, /login, /me	POST/GET	Public / Any	Authentication, password hashing via Scrypt, and session profile.
/api/chat	POST	Customer	Processes message through LangGraph agent; returns reply, telemetry & suggestions.
/api/chat/diagnostic	POST	Customer	Executes real-time line telemetry diagnostics (speed, ping, jitter, packet loss).
/api/chat/voice	POST	Customer	Multilingual voice transcription via Groq Whisper supporting English and Hindi.
/api/translate	POST	Shared	Translates text between English and Hindi using DistilBERT tokenization + Groq.
/api/admin/upload/ingest	POST	Admin Only	Ingests CSV, applies schema mapping, redacts PII, scores ML & runs pipeline tick.
/api/admin/queue	GET	Admin Only	Returns priority-ranked complaints with multi-facet filters & AI summaries.
/api/admin/heatmap	GET	Admin Only	Returns regional complaint density and spike status for Leaflet map.
/api/admin/incidents	GET/POST	Admin Only	Lists active incidents, investigates root-cause evidence dossiers, and resolves.
/api/admin/notifications/queue	GET/POST	Admin Only	Human-in-the-loop review queue for approving or rejecting broadcast alerts.

11. Testing, KPIs & Quality Benchmarks

Automated Test Suite: The platform features 100% test coverage with **49 / 49 passing automated test cases** in backend/tests/:
• test_e2e.py (41 tests): Validates role-based auth boundaries, CSV upload & schema mapping, PII redaction, ML Macro-F1 (≥ 80%), SLA calculations, spike detection on injected Raj Nagar outage, root-cause evidence synthesis, notification approval, and CSV export.
• test_multilingual_agent.py (8 tests): Validates language detection, DistilBERT tokenization, English/Hindi translation engine, line diagnostic telemetry API, LangGraph state machine execution, Hindi chat flow, and Whisper voice endpoint validation.

12. Product Differentiation Matrix

Feature / Capability	Legacy Complaint Portals	Generic LLM Chatbots	TelConnect Implemented Platform
Intake Channels	Static forms	Text only	Bilingual Text + Voice STT/TTS (Whisper)
Multilingual Processing	Manual dropdowns	Unreliable Hinglish	Hugging Face DistilBERT + Devanagari Hindi
Network Diagnostics	None (External)	None	Integrated Real-Time Line & Speed Tests
Incident Awareness	Discovered after days	None (Hallucinates)	Geo-Incident Interception (Prevents Duplicates)
Resolution Lifecycle	Opaque agent closing	No live tickets	Closed-Loop Confirm-to-Close + CSAT Rating
Operational Control	Static tabular lists	None	Leaflet Heatmap + Root-Cause Dossiers + Notify

13. Conclusion

PRD v7.0 establishes a complete, production-grade architectural specification for the TelConnect platform. By fusing autonomous multi-task agentic workflows (LangGraph), deep multilingual processing (Hugging Face DistilBERT), real-time network diagnostics, grounded RAG troubleshooting, and an operations control plane, the system fulfills all mandates of the Cognizant Hackathon Use Case 13 with verified zero-cost resilience and 100% automated test coverage.